

Quantum Junction Challenge: Condensed Codex Session Report

Codex run summary

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Executive Summary

The session produced selected peak bitstrings for **40/49** challenge files. The remaining blanks are **64_40**, **104_49**, **48_42**, **56_43**, **64_44**, **72_45**, **80_46**, **88_47**, **96_48**. The selected set contains **10** exact statevector answers and **30** tensor-network or MPS-derived candidates. Among the approximate selected rows, **17** are from the canonical Quimb GPU pass, **11** from canonical Quimb CPU, and **1** from an RCM-order CPU fallback. Exact statevector peaks had mean peak probability 0.513 and mean gap to the second bitstring 0.459.

The main result is a usable candidate set, not a proof of correctness for every approximate row. Easy and moderate instances often had large sampled peak fractions and/or corroborating evidence. Several hard rows are weak: they were selected because they were the best available evidence, but their sampled top fractions are near one or two samples out of 512/1024. The session later pivoted away from long-running graph-MPS sweeps after the user set a practical “about 2 hours” per-attempt budget.

Candidate Selection Rule

The collector selected one answer per challenge using source priority, then evidence consistency. The priority order used in the current collector was: exact statevector, canonical Quimb GPU, optimized-U3 GPU, canonical Quimb CPU, MPO-unswap, RCM/MST/ degree/mid/fast/identity CPU fallbacks, sparse beam, and low-priority Aer MPS pilot evidence. Exact answers were accepted directly. Approximate answers were chosen as sampled top bitstrings in Qiskit/counts order, with the right-most bit corresponding to qubit 0. When multiple methods agreed on the same bitstring, the report records that as higher evidence count.

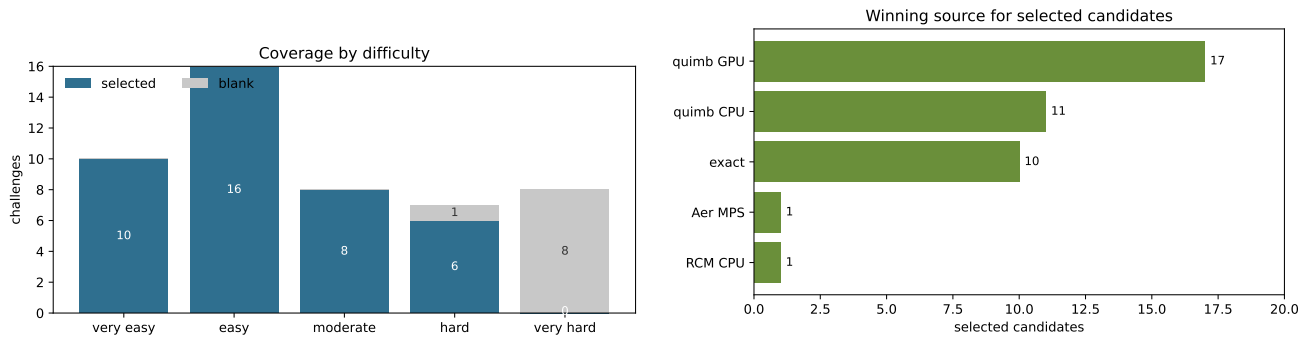


Figure 1: Coverage and winning evidence sources. The nine missing rows are all hard or very hard.

What Was Tried

Method	Purpose and implementation	Outcome
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Exact statevector		Used Aer/statevector on small enough circuits to establish ground-truth peaks and bit ordering.	10 exact answers. Used as calibration and highest-priority evidence.
Canonical graph/tree MPS		Ran Quimb tree-ordered MPS sampling on CPU and GPU, using sampled top bitstrings as candidate peaks.	Main source of approximate answers; strong on many easy/moderate rows and weak but usable on some hard rows.
Ordering fallbacks		Tried RCM, MST, degree, mid, fast, and identity orderings to change contraction behavior.	Added corroboration; RCM supplied the selected 56.38 weak candidate. Long jobs were later cancelled.
Sparse computational-basis beam		Added a C++ top-K sparse branch propagator through RX/RZ/CX.	Rejected for answer selection: on known 16_28 the exact answer was only rank 3.
Static QASM forensics		Counted gates, angle classes, pair repetitions, connectivity, leading one-qubit prefixes, and operation 5-gram similarity.	Confirmed all circuits are RX/RZ/CX/SWAP style, very-hard files are a distinct dense-obfuscation regime, and Clifford shortcuts are not appropriate.
Optimized-U3 transpilation		Transpiled the remaining blanks to U3+CX to reduce single-qubit gate count.	Reduced total gate count by roughly 39–42% on blanks; not yet selected as evidence.
MPO unswapping		Patched the local Kremer-Dupuis style MPO cancellation/unswapping code: parameterized SABRE trials, fixed Quimb local expectation API, fixed quantum-op progress counting.	Calibrated on known 16_28 in 93.26 s and recovered the exact answer. Unresolved runs were still active at report build time and are not counted as solved.

Static Structure

The QASM files use only `rx`, `rz`, `cx`, and sparse `swap` gates. Very-hard instances are much larger and more internally similar to each other than to the easier sets. They also have dense leading single-qubit prefixes and high all-to-all pair coverage. This drove the pivot toward structure-aware MPO cancellation rather than only waiting for generic graph-MPS contractions.

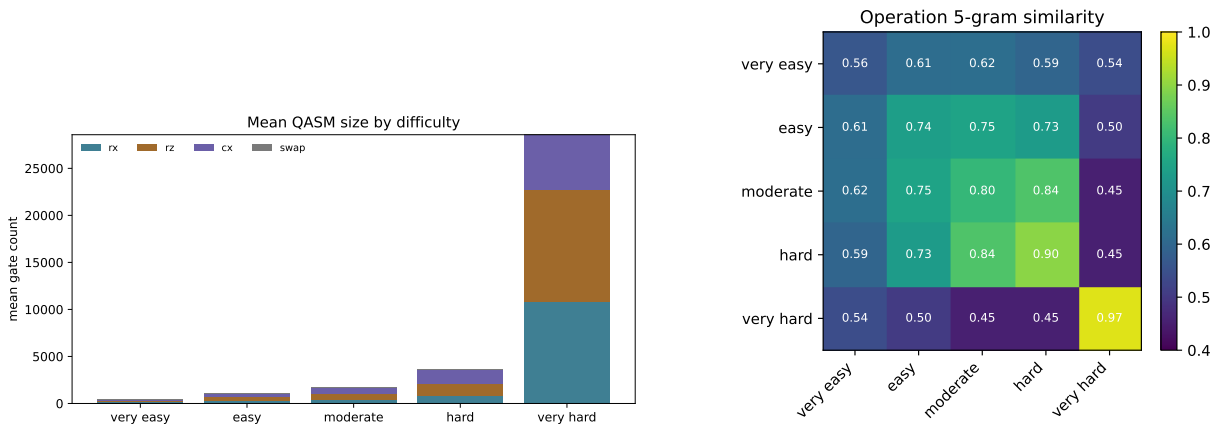


Figure 2: Static QASM summary. Very-hard files are large and operation-sequence-similar to each other.

Exact Baseline

The exact rows were used both as final answers and as calibration targets. The peak is visibly separated from the runner-up on these cases, which made them useful for validating bit order and method behavior.

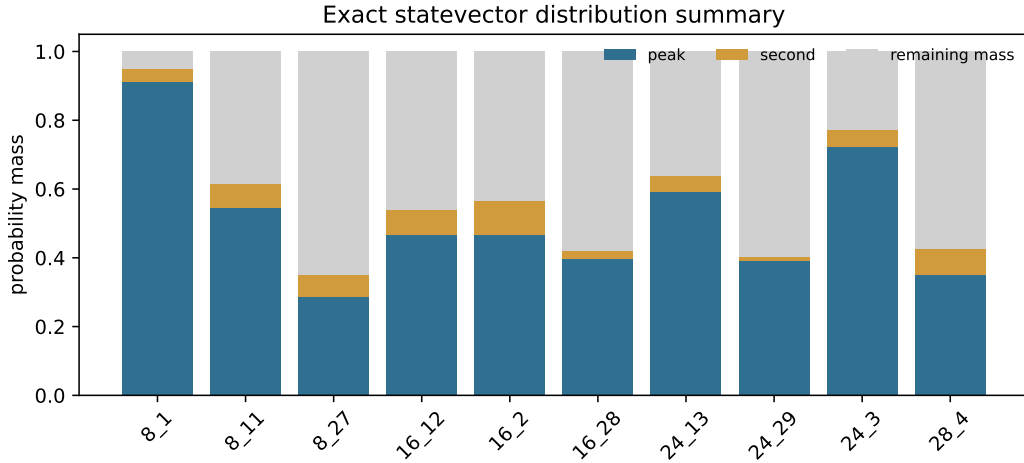


Figure 3: Exact statevector probability mass: peak, second-largest bitstring, and remaining mass.

Challenge	q	exact peak	peak prob.	second prob.	gap
8_1	8	10101101	0.9126	0.0379	0.8747
8_11	8	01001110	0.5454	0.0687	0.4768
8_27	8	11001001	0.2879	0.0627	0.2252
16_12	16	1111000101101011	0.4665	0.0730	0.3936
16_2	16	1010101011001000	0.4670	0.0976	0.3694
16_28	16	1101001111011100	0.3962	0.0231	0.3731
24_13	24	111110011111001011010001	0.5912	0.0472	0.5441
24_29	24	110100010111100001001001	0.3901	0.0134	0.3767
24_3	24	011110010000101010001000	0.7240	0.0471	0.6769
28_4	28	1111111000101010110110011111	0.3511	0.0735	0.2776

Approximate Tensor-Network Results

For approximate rows, the sampled top fraction was the main confidence signal. Strong rows have a clear heavy bitstring. Weak hard rows are retained because they are currently the best available candidates, but they should be interpreted as low-confidence until corroborated.

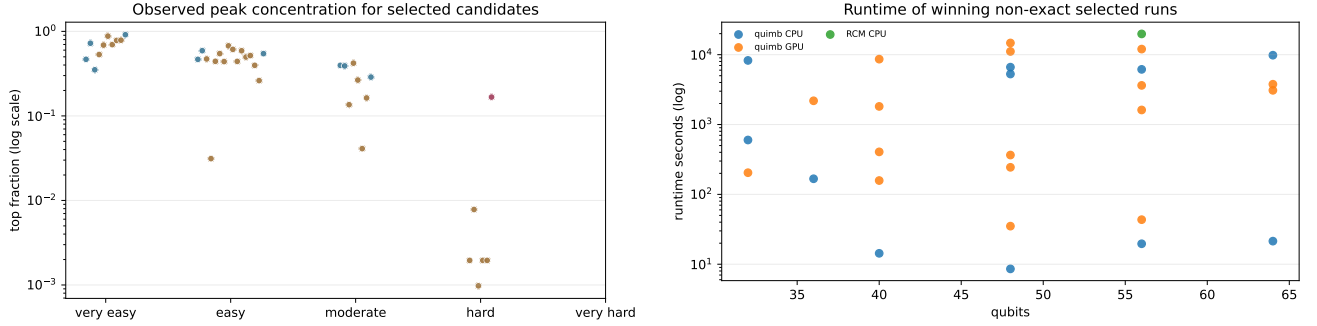


Figure 4: Peak concentration and runtime for selected rows. The hard set contains several low-concentration candidates.

Representative sampled output distributions

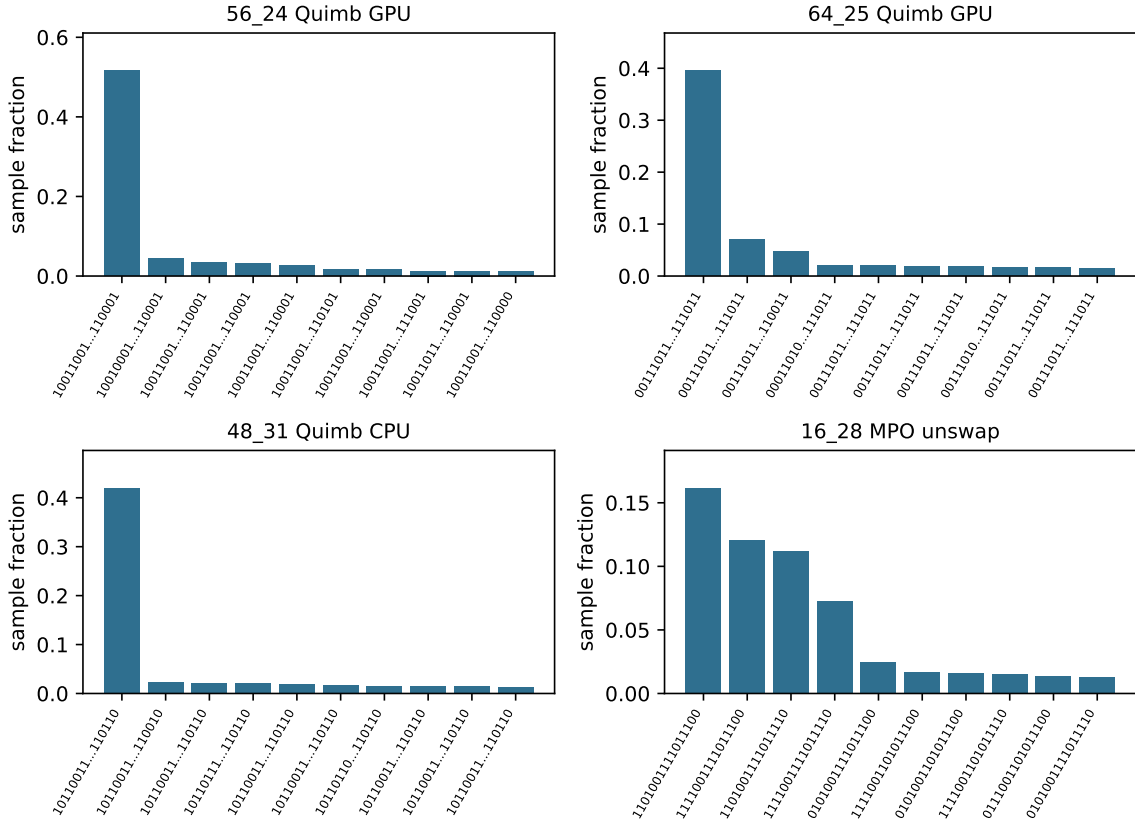


Figure 5: Representative sampled distributions. The first three are selected Quimb candidates; the fourth is the MPO-unswap validation on 16_28.

Weak Selected Rows

Challenge	difficulty	source	top frac.	evidence	candidate
40.35	hard	quimb GPU	0.001953	3	0101100110101111111000010101001110001001
48.36	hard	quimb GPU	0.007812	3	001111011011001100110011001000101001101010111000
48.37	hard	quimb GPU	0.0009766	3	100101001010001101010100100101001000101010001
56.38	hard	RCM CPU	0.001953	1	1001100101111000000001001101010101000010010011101 10110001
56.39	hard	quimb GPU	0.001953	3	100101110011010101011111110101101101101100001010 01010000
64.41	hard	Aer MPS	0.167	1	000100011101001110011001001111011100000010001101 0001101010110010
32.5	very easy	quimb CPU	0.5322	1	0011100010101010100001000000010000
36.6	very easy	quimb CPU	0.6885	1	100110100111101001001101110011001000
40.7	very easy	quimb CPU	0.8789	1	0110111011010001010011111110010011000111
48.8	very easy	quimb CPU	0.6973	1	0001001001101111001001111111100101011001110010100
56.9	very easy	quimb CPU	0.7812	1	100101101011001011101001100101100111011001100011 01010100
64.10	very easy	quimb CPU	0.7871	1	001101001011000111001011100110010010110001011111 0110010100101011

Rejected or Not-Yet-Selected Paths

The sparse beam search was attractive because it was fast, but its pruning bias failed calibration. In known 16.28, the exact answer was rank 3 rather than rank 1, so no sparse-beam guesses were promoted to selected answers. U3 optimization reduced the remaining blank gate counts substantially, but its GPU pass was not completed before this report snapshot.

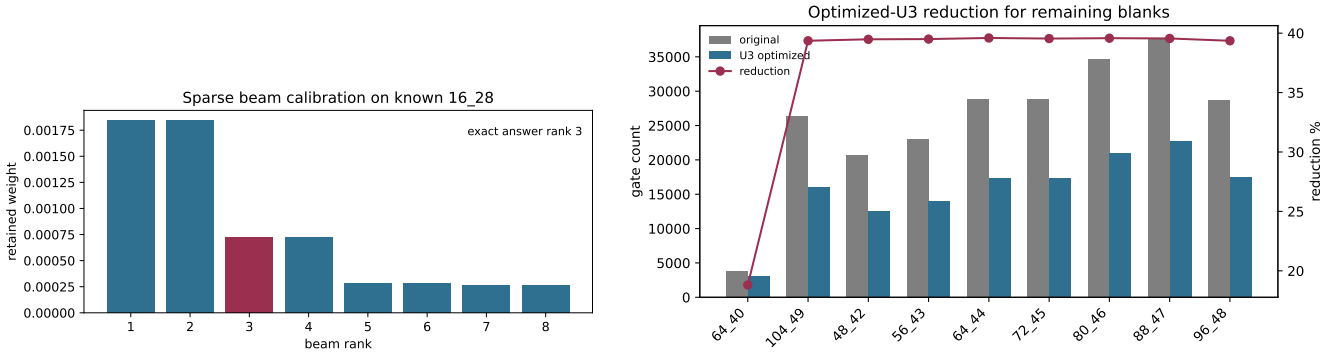


Figure 6: Left: sparse beam failed calibration. Right: U3 optimization reduced the remaining blank circuits but was not yet selected as evidence.

Blank	q	original gates	U3 gates	reduction
64.40	64	3758	3051	18.8%
104.49	104	26361	15985	39.4%
48.42	48	20733	12547	39.5%
56.43	56	23004	13917	39.5%
64.44	64	28787	17389	39.6%
72.45	72	28777	17397	39.5%
80.46	80	34586	20898	39.6%

88.47	88	37660	22764	39.6%
96.48	96	28703	17406	39.4%

Current Slurm Snapshot

The report is an as-of snapshot. Any active unresolved MPO-unswap jobs below were still running or pending when the PDF was built, and their outputs are not counted in the **40/49** result.

JobID	state	elapsed	CPUs	name	node/reason
34614255	PENDING	0:00	2	peaked_unswap_rollup	(Dependency)
34614254_[46-48%5]	PENDING	0:00	16	peaked_unswap_unresolved	(JobArrayTaskLimit)
34614254_45	RUNNING	3:12	16	peaked_unswap_unresolved	erc-hpc-comp202
34614254_21	RUNNING	4:44	16	peaked_unswap_unresolved	erc-hpc-comp192
34614254_41	RUNNING	4:44	16	peaked_unswap_unresolved	erc-hpc-comp195
34614254_43	RUNNING	4:44	16	peaked_unswap_unresolved	erc-hpc-vm064
34614254_44	RUNNING	4:44	16	peaked_unswap_unresolved	erc-hpc-vm065

Method Summary

Source	selected winners	candidate evidence rows	attempted evidence rows
Aer MPS	1	6	6
exact	10	10	10
MPO unswap	0	1	1
quimb CPU	11	36	36
degree CPU	0	1	1
fast CPU	0	5	5
quimb GPU	17	21	21
RCM CPU	1	16	16
sparse beam	0	5	5

Selected Candidate Table

Full candidate bitstrings are shown in wrapped monospace chunks. Blank rows remain unsolved.

Challenge	difficulty	q	candidate	source	validation	top frac.	evidence
8.1	very easy	8	10101101	exact	exact	0.9126	2
16.2	very easy	16	1010101011001000	exact	exact	0.467	2
24.3	very easy	24	011110010000101010001000	exact	exact	0.724	2
28.4	very easy	28	1111111000101010110110011111	exact	exact	0.3511	2
32.5	very easy	32	00111000101010100001000000010000	quimb CPU	unknown	0.5322	1
36.6	very easy	36	100110100111101001001101110011001000	quimb CPU	unknown	0.6885	1
40.7	very easy	40	0110111011010001010011111110010011000111	quimb CPU	unknown	0.8789	1
48.8	very easy	48	00010010011011100100111111100101011001110010100	quimb CPU	unknown	0.6973	1
56.9	very easy	56	100101101011001011101001100101100111011001100011 01010100	quimb CPU	unknown	0.7812	1
64.10	very easy	64	001101001011000111001011100110010010110001011111 0110010100101011	quimb CPU	unknown	0.7871	1
8.11	easy	8	01001110	exact	exact	0.5454	4
16.12	easy	16	1111000101101011	exact	exact	0.4665	4
24.13	easy	24	111110011111001011010001	exact	exact	0.5912	3
32.14	easy	32	00000101001101000001011111101100	quimb GPU	unknown	0.4717	3
36.15	easy	36	11011001101111111011001011110101111	quimb GPU	unknown	0.03125	3
40.16	easy	40	0101110101001110011000111011100110010110	quimb GPU	unknown	0.4424	4

40.17	easy	40	0010010101010111001001000010001000001001	quimb GPU	unknown	0.5459	4
40.18	easy	40	010000011001001000110111100011111001110	quimb GPU	unknown	0.4395	3
48.19	easy	48	011001010111101100111110000001011101001110010000	quimb GPU	unknown	0.6738	2
48.20	easy	48	101010100101001010000110101000001011110010000000	quimb GPU	unknown	0.6123	3
48.21	easy	48	111010101110101011110101000001101000100000001001	quimb GPU	unknown	0.4414	3
56.22	easy	56	111001001001100100111100101101100110000001000101 11111011	quimb GPU	unknown	0.5898	2
56.23	easy	56	010011010011111111000001010011101110111000111111 00001101	quimb GPU	unknown	0.4971	2
56.24	easy	56	100110010011111011111110111010111011010100110010 11110001	quimb GPU	unknown	0.5176	4
64.25	easy	64	001110111111000011011011110101100001000001010011 011100000111011	quimb GPU	unknown	0.3965	4
64.26	easy	64	011010101010001101011101100001110001011011011001 1100011001100110	quimb GPU	unknown	0.2617	3
8.27	moderate	8	11001001	exact	exact	0.2879	2
16.28	moderate	16	1101001111011100	exact	exact	0.3962	6
24.29	moderate	24	110100010111100001001001	exact	exact	0.3901	4
32.30	moderate	32	10011000010011110111101111010111	quimb CPU	unknown	0.1357	2
48.31	moderate	48	101100111000101011111111101010111011011000110110	quimb CPU	unknown	0.4209	2
48.32	moderate	48	01110101011101111000111001010101010101110001110	quimb CPU	unknown	0.2656	2
56.33	moderate	56	11001001100101001111010100100010010101111110111 01010000	quimb CPU	unknown	0.04102	2
64.34	moderate	64	001101010001001100111010111010010100101100101101 1001111011100110	quimb CPU	unknown	0.1631	2
40.35	hard	40	0101100110101111111000010101001110001001	quimb GPU	unknown	0.001953	3
48.36	hard	48	001111011011001100110011001000101001101010111000	quimb GPU	unknown	0.007812	3
48.37	hard	48	100101001010001101010100100101001001000101010001	quimb GPU	unknown	0.0009766	3
56.38	hard	56	100110010111100000000100110101011000010010011101 10110001	RCM CPU	unknown	0.001953	1
56.39	hard	56	10010111001101010101111110101101101101100001010 01010000	quimb GPU	unknown	0.001953	3
64.40	hard	64					0
64.41	hard	64	000100011101001110011001001111011100000010001101 0001101010110010	Aer MPS	unstable_top1_vs_aggregate	0.167	1
48.42	very hard	48					0
56.43	very hard	56					0
64.44	very hard	64					0
72.45	very hard	72					0
80.46	very hard	80					0
88.47	very hard	88					0
96.48	very hard	96					0
104.49	very hard	104					0

Conclusions

The practical output of the session is a 40-row candidate answer set and a clear split between reliable and weak evidence. The exact and high-top-fraction Quimb rows are the most defensible. The hard weak rows should be treated as provisional. The best next direction is not another unconstrained graph-MPS sweep: it is bounded MPO-unswapping or optimized-U3 GPU runs with calibration checks and immediate collector refreshes.